

MECHANICAL PROPERTIES OF MATTER

Advanced Level Science for Technology (SFT) - TechHub LK Study Guide

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1. Elasticity

The property of a body by virtue of which it tends to regain its original shape and size after the removal of the deforming force is called **elasticity**.

Objects that exhibit this property are known as **elastic bodies**.

- **Examples of Elastic Bodies:** Glass, Wood, Iron, Rubber, Various metals [cite: 4].

Objects that do not show elasticity and retain their deformed shape after the deforming force is removed are called **inelastic (plastic) bodies**.

- **Examples of Inelastic Bodies:** Clay, Putty, Paper [cite: 4].

2. Extension (Strain/Deformation - e)

The change in length of an elastic object resulting from an applied external force is known as **extension**.

- **Units of Measurement:** meters (m), centimeters (cm), millimeters (mm) [cite: 4].

3. Strain

Strain is defined as the ratio of the extension (e) to the original length (l) of the object [cite: 4].

$$\text{Strain} = \frac{\text{Extension } (e)}{\text{Original Length } (l)}$$

Note: Strain is a dimensionless quantity and has no units [cite: 4].

4. Stress

Stress is defined as the restoring force per unit area, or the internal resistance force set up per unit cross-sectional area of a deformed elastic body [cite: 4].

$$\text{Stress} = \frac{\text{Force } (F)}{\text{Area } (A)}$$

- **Units of Stress:** N / m^2 or $\text{N} \text{ m}^{-2}$ [cite: 4].

Types of Stress

1. **Tensile Stress:** The stress set up when a stretching (tensile) force is applied to an elastic object to increase its length [cite: 4].

$$\textit{ext{Tensile Stress}} = \textit{rac{ ext{Tensile Force}}{ ext{Cross-sectional Area}}}$$

2. **Compressive Stress:** The stress set up when a compressive force is applied to an elastic object to decrease its length [cite: 4].

$$\textit{ext{Compressive Stress}} = \textit{rac{ ext{Compressive Force}}{ ext{Cross-sectional Area}}}$$

5. Hooke's Law

Hooke's Law states that within the limit of proportionality, the applied external force on an elastic object is directly proportional to the extension produced [cite: 4].

$$F \propto e \implies F = K e$$

Where **K** is a constant depending on the elastic object, known as the **spring constant** for springs, and the **force constant** for wires and rods [cite: 4].

From the relation: $K = \textit{rac{F}{e}}$, the SI unit of **K** is N m^{-1} [cite: 4].

6. Force vs. Extension Graph & Behavior

Plotting the external force against extension for an elastic wire yields a straight line passing through the origin (within the proportional limit). Key points on the loading/deformation curve include [cite: 4]:

- **Proportional Limit:** The point up to which Hooke's law is strictly obeyed.
- **Elastic Limit:** Beyond the proportional limit, the material reaches its elastic limit. If the load is removed here, the object returns to its original length without permanent set.
- **Brittle Materials:** Brittle materials tend to fracture soon after the proportional limit without significant plastic deformation [cite: 4].
- **Fracture Point:** The point where the material breaks under sustained loading.

7. Young's Modulus (\$Y\$)

Within the limit of proportionality, the ratio of tensile stress to tensile strain is defined as **Young's Modulus** [cite: 4].

$$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{F / A}{e / l}$$

- **Units:** N m^{-2} [cite: 4].
- **Rearranged Form:** From $Y = \frac{F}{Ae}$, we get $e = \left(\frac{l}{A} \right) F$, which takes the straight-line equation form $y = mx$ where slope $m = \frac{l}{A}$ [cite: 4].
- **Applications:** Young's modulus varies depending on the type of matter and is crucial when selecting wires, rods, and calculating safe load capacities for concrete pillars in construction [cite: 4].

8. Elastic Potential Energy

The energy stored in an elastic object due to deformation by an external force is called **elastic potential energy**. The work done on the object is equal to the potential energy stored [cite: 4].

$$\text{Elastic Potential Energy} = \text{Work Done} = \text{Average Force} \times \text{Extension}$$

$$\text{Potential Energy} = \frac{0 + F}{2} \times e = \frac{1}{2} F e$$

Substituting Hooke's Law ($F = Ke$) into the equation [cite: 4]:

$$\text{Potential Energy} = \frac{1}{2} \times (Ke) \times e = \frac{1}{2} K e^2$$

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